Gabriele Novello

Graphics programmer with experience in Vulkan and real-time rendering systems.

Turin, Italy
+39 388 792 6877
g.novello1006@gmail.com
frol3z.github.io/portfolio

Education

University of Turin 2024 – 2026
M.Sc., Computer Science Turin, Italy

- final grade: 110/110 cum laude, with distinction and recommendation for publication
- part-time position: translated computer graphics course slides from Italian to English
- master's thesis: a probe-based framework for dynamic real-time global illumination
- major topics studied: computer graphics, virtual reality systems, image processing & computer vision, parallel & distributed systems, machine learning & deep learning

University of Turin 2021 – 2024
B.Sc., Computer Science Turin, Italy

- final grade: 110/110 cum laude
- bachelor's thesis: game state saving approaches in gaming
- developed a small roguelite game in Godot to test the methods proposed in the thesis

Personal Projects

Lattice-Boltzmann fluid solver 2026
Parallelization and performance analysis

- parallelized C implementation with OpenMP and OpenMPI
- analyzed scalability on a cluster of 36 nodes using Slurm as a workload manager

Felina 2025 – Present
Real-time Vulkan Renderer

- built with C++, modern Vulkan (1.3+), CMake and HLSL
- features PBR materials, deferred shading and multiple punctual lights support
- includes Dear ImGui user-interface and scene loading through the glTF file format

Monte Carlo Denoiser 2024 – 2025
Standalone Tool / Research

- built and trained a ResNet-based neural network using PyTorch to denoise Monte Carlo renderings using Disney's published dataset and methodology

Technical Skills

Programming & APIs

C/C++ · Python · Java · C# · Vulkan · Slang · HLSL · CUDA · OpenMP · MPI

Tools

Git · RenderDoc · CMake · Visual Studio · Godot · Unreal Engine

Languages

- English · proficient
- Italian · native
- Japanese · elementary

Interests

- cinematography, filmmaking and storytelling
- digital and film photography
- video games and game development
- classic and electric guitar, music production
- learning: computer science, physics and foreign languages